PROJECT TITLE: HUMAN RESOURCES, PAY ROLL AND RECRUITMENT AUTOMATION

1. ABSTRACT

In many organizations, HR operations, recruitment workflows, payroll calculations, and project tracking are handled separately using paper registers, email chains, messenger, and Excel sheets. This manual workflow wastes time, causes mistakes in salary calculations, loses track of candidate resumes, misses unpaid leaves, and makes it hard for managers to know the real daily progress of ongoing company projects. In this project, I have developed an Automated Human Resource, Recruitment, Payroll, and Project Tracking Management System using Python Django, MySQL, and a Bootstrap frontend. The system centralizes hiring and employee records, tracks daily attendance, manages leaves, automates monthly salary disbursement, and includes an active Project Progress Tracking module.

The system provides an integrated architecture featuring a public Careers portal alongside 3 role-based portals: Admin, HR Manager, and Employee. Job seekers can view open vacancies and apply online with their resumes. HR managers track candidates through a structured recruitment pipeline: screening resumes, scheduling interview rounds, logging evaluation feedback, and converting selected candidates into registered employees with one click. For existing staff, HR onboards employees, uploads documents, creates company projects, assigns them to employees, and monitors project completion in real time. Employees log in to update their project progress percentage, submit work remarks, view their profile, track leave balances, apply for leaves, and clock in/out with Office Wi-Fi IP verification to prevent fake attendance. The system also features a Payroll Automation Engine: with one click, HR calculates monthly salaries with automatic Loss of Pay (LOP) for absences, deducts PF and taxes, and generates official downloadable PDF pay slips. This makes company operations organized, productive, and paperless.

2. OBJECTIVES:

1. To publish company job openings on an online careers portal and accept candidate applications with resume uploads.
2. To manage the end-to-end interview process, enabling HR to shortlist applicants, schedule interview rounds, record interviewer feedback, and convert selected candidates directly into employee records with a single click.
3. To store employee records, departments, designations, bank details, and uploaded documents in a single, secure MySQL database.
4. To enable online clock-in and clock-out with automatic office Wi-Fi IP validation so employees cannot mark proxy attendance or clock in from home.
5. To let employees apply for leaves online, automatically track remaining leave balances, and allow HR to approve or reject them.
6. To automatically calculate gross pay, Loss of Pay (LOP) for absent days, provident fund, and taxes without manual Excel math.

7. To allow HR to process company-wide monthly payroll in one click.
8. To automatically generate official, downloadable PDF salary slips using ReportLab.
9. To separate permissions between Admin, HR, and Employees so sensitive financial and recruitment data remains confidential.
10. To enable HR to create and assign projects to employees, and allow employees to update daily progress percentages and completion remarks.

3. EXISTING SYSTEM:

Currently, companies manage HR, hiring, payroll, and project updates using disconnected tools: paper attendance books, Excel files, email inboxes for resumes, and verbal or WhatsApp status updates.

LIMITATIONS:

1. Resumes get lost in messy email inboxes. Interview scheduling and feedback notes are tracked on paper. When a candidate is hired, HR must manually re-type all their personal information into employee registers, causing double work and typos.
2. Managers must constantly call or message employees on messenger to ask for project updates, leading to lost details and missed deadlines.
3. Manual calculation of basic pay, allowances, overtime, and deductions in Excel frequently causes salary errors.
4. HR spends days manually cross- referencing attendance registers with Excel salary sheets.
5. If an employee takes unapproved leave and HR forgets to note it down, the employee gets paid for days they didn't work.
6. Traditional registers and basic web forms allow employees to mark proxy attendance or clock in from home.
7. Employees have no single dashboard to view their salary slips, apply for leaves, or report daily work progress.

4. PROPOSED SYSTEM

The proposed system is a secure, web-based platform that replaces spreadsheets and paper registers with database-driven automation.

How System Works:

1. Online Job Postings & Public Careers Page: HR creates and publishes open job positions with requirements and vacancy counts. External candidates visit the careers page, view job descriptions, and apply by submitting their personal details and resumes.
2. Structured Interview Pipeline & 1-Click Hiring: HR reviews applications, shortlists promising candidates, and schedules interview dates (Technical / HR rounds). Interviewers log feedback and ratings. When a candidate passes and is marked "Selected", HR clicks "Convert to Employee", which automatically creates their official profile in the employee database without re-typing information.

3. Attendance with Wi-Fi IP Check: When an employee clicks "Clock In", the system checks their IP address. It only marks them "Present" if connected to the Office Wi-Fi; otherwise, it blocks the attempt.
4. Leave Management & Auto-Deductions: Employees apply for leaves online. When HR approves a leave, the employee's remaining leave balance updates automatically.
5. Project Progress Tracking: HR creates projects, sets deadlines, and assigns team members. Employees log in to view their assigned projects, update completion percentages (e.g., 25%, 50%, 100%), and write daily status notes.
6. Automated Loss of Pay (LOP): The system calculates daily basic pay and it multiplies this by unapproved absent days and deducts it from the gross pay automatically.
7. Statutory Deductions: Automatically applies 12% Provident Fund (PF) and income tax rules.
8. One-Click Monthly Payroll: HR selects the month (e.g., 2026-10) and clicks "Process Payroll". The engine computes all employee salaries and stores them in MySQL within seconds.
9. Instant PDF Pay slips: Both HR and employees can click "Download PDF" to get a formatted, official salary slip anytime.

5. ROLES AND RESPONSIBILITIES

1. Admin:

- Sets up the core system, departments, and designations.
- Creates user accounts for multiple HR managers.

2. HR Manager

- Posts and closes job vacancies, reviews incoming candidate applications, and downloads resumes.
- Schedules interview dates, records interviewer evaluation feedback, and updates candidate stages.
- Converts selected candidates into active employees and generates login credentials.
- Views and edits employee records, departments, and designations.
- Creates projects, assigns them to employees, and monitors progress bars and status updates.
- Monitors daily attendance, reviews leave requests, and approves or rejects them.
- Executes the payroll engine, reviews company-wide payout summaries, and downloads individual or batch PDF pay slips.

3. Employee:

- Logs in using their credentials and can change their initial password.
- Views their personal profile, department, and salary breakdown.
- Clocks in and clocks out daily with Office IP verification.
- Updates daily project progress percentages and add’s completion remarks.

- Views remaining leave balances, submits leave requests, and tracks approval status in real time.
- Downloads monthly PDF pay slips without needing to contact HR.

4. Job Seeker / Candidate:

- Browses active job openings by department and role.
- Submits job applications online along with resume uploads.
- Tracks application status and views scheduled interview details
- Attend scheduled interview rounds and receives selection status updates.

6. TECH STACK

Components	**Technology**
Back-end	Python Django
Database	MySQL
Database connector	mysqlclient
Front-end	HTML, CSS, JavaScript, bootstrap
Libraries	reportlab
Security	Bcrypt, Django session Authentication